

Assignment 2: Tic-Tac-Toe

due Monday 07/06 at 9:00am (EST)

After finishing this assignment, submit the `assignment-2.py`, `assignment-2.ipynb`, and `assignment-2-logs.txt` files on Canvas.

Introduction

Let's recap!

In general, our assignments will consist of questions where AI usage is not allowed and those where AI use is expected. Questions where AI use is expected will be explicitly marked as such; if a question has no such label, assume that AI use is not permitted.

You will usually be expected to submit three files on Canvas for every assignment. These are: a `.py` file, where you will write any required code, a `.txt` file with any AI chatbot conversations related to the assignment and any other outside sources you use, and a `.ipynb` file that answers any written questions associated with the assignment.

For this assignment, as with all other work for this course, you are expected to follow the academic integrity guidelines in the course syllabus and Cornell's [Code of Academic Integrity](#).

General Tips + Guidelines

There are several tips and guidelines that you should follow for this assignment:

- You will want to run the code in `assignment-2.py` from the terminal. You can use VSCode's terminal or your computer's terminal app.
- You should run your code in the virtual environment we created in Lab 1. If you have trouble figuring out how to activate it, try Googling! If you're still stuck, reach out to me – the earlier the better.
- Run your code regularly as you write it. It's much easier to catch errors if you debug early and often! I highly recommend that you run the included tests for each task directly after writing the function they test.
- The assignment is broken down into many tasks and it's easy to skip one by accident. Before submitting, go through each task to make sure you've completed it.
- For this assignment, you shouldn't use, and don't need, any Python constructs not yet covered in class. You should demonstrate your skills with the Python we've learned so far!

Part 0

For this assignment, we'll be building a game of tic-tac-toe (that you'll eventually be able to play!).

Before beginning, make sure that you have downloaded the `assignment-2.pdf` instructions, `assignment-2.py`, and `assignment-2.ipynb` files. These files should be placed within the `info1100/assignments` folder.

All code for this assignment will be written in `assignment-2.py`.

Task 1: Fill in the `# Author` and `# NetID` comments at the top of the `.py` file.

Part 1

Once again, the `assignment-2.py` file already has some code in it! It contains a stub for each function you'll write, two helper functions that let you play the game at the end (`player_2_turn` and `play_game`), and — under the `if __name__ == '__main__':` block — a series of tests you can uncomment to check your code as you go.

For now, you don't need to worry about how the code in the `player_2_turn` and `play_game` functions works. You also won't *necessarily* need to understand how the tests work, but they're written using only Python elements we've already learned, and understanding how they work will probably help you debug!

As you work through the tasks in this section, you should uncomment the provided tests. The tests for each task are preceded by a header with the task number; once you're ready to test your function, uncomment the related section of tests. Each test is of the form `'[function output] should be [expected output]'`. If the `[function output]` and `[expected output]` sections match, your code is working! If not, check out what tests are failing, and try to debug your code from there. As always, you should come to me if a genuine effort to debug your code hasn't gotten you unstuck!

Task 2: Write the `get_cell_string` function according to the description in the function comment and run the associated tests.

Task 3: Write the `print_row` function according to the description in the function comment and run the associated tests.

Task 4: Write the `print_board_state` function according to the description in the function comment and run the associated tests.

Task 5: Write the `is_valid_grid_cell` function according to the description in the function comment and run the associated tests. Note that this function already has some code in it; the code uses some syntax you haven't learned yet to ensure that `x` and `y` are (or can be converted into) integers.

Task 6: Write the `update_board_state` function according to the description in the function comment and run the associated tests.

Task 7: Write the `player_row_win` function according to the description in the function comment and run the associated tests.

Task 8: Write the `player_column_win` function according to the description in the function comment and run the associated tests.

Task 9: Write the `player_diagonal_win` function according to the description in the function comment and run the associated tests.

Task 10: Write the `is_win` function according to the description in the function comment and run the associated tests.

Part 2 (AI Tasks)

These tasks will all ask you to use an AI chatbot. You may select the AI chatbot of your choice. You should copy and paste every conversation you have with a chatbot related to this assignment into a text file called `assignment-2-logs.txt`. You're expected to understand on a high level any code you include in the assignment; you will be asked to describe all AI-generated code. You may also be asked to explain or edit this code in oral assessments.

All written questions should be answered in the `assignment-2.ipynb` file.

Task 11: In the `assignment-2.ipynb` file, fill in the blanks that ask for your name, NetID, and which AI chatbots you use. If you want to use multiple chatbots, specify which one is used for which question!

Task 12: Writing the `is_draw` function will require some Python syntax we haven't learned yet. Use an AI chatbot to fill in the function according to the provided comment and paste your generated code into the function body. Then, make sure it's working by running the associated tests.

Task 13: Continue prompting the chatbot until you have a general understanding of the code it wrote for Task 12. Then, under the Task 13 header in the `assignment-2.ipynb` file, write a description of the Task 12 code.

With this final function filled in, you should be able to play a game of tic-tac-toe! Uncomment the final section of code under the `Play the game!` header that calls `play_game`. Battle the computer to see who wins!